

“ IF THE **SPINE** OF A STEVEDORE [DOCKWORKER]... **SUPPORTS** A LOAD OF **120 POUNDS**... THE **FORCE OF NATURE** EXERTED **IN THE DISK** AND... **MUSCLES** OF THE SPINE IS EQUAL TO **25,585 POUNDS**. ”

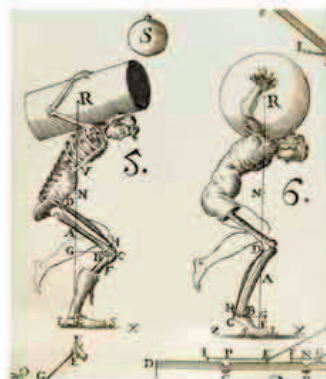
Giovanni Borelli, Italian physiologist, 1670s



GERMAN MATHEMATICIAN Gottfried Leibniz had investigated a **binary number system** in which numbers are represented by just two symbols: 0 and 1. In 1679, he suggested using such a system as the **basis for a computing machine**.

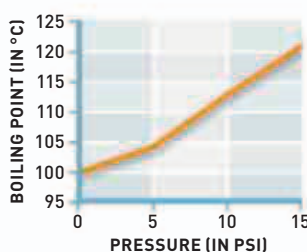
This year, French inventor **Denis Papin** (1647–1712) collaborated with his Anglo-Irish counterpart Robert Boyle on a **steam digester**. This cooking device using high-pressure steam made it possible to extract fat from bones, and led to the development of the steam engine and pressure cooker. Italian physiologist and physicist **Giovanni Borelli** (1608–79) spent much of

Papin's steam digester
The safety valve that Papin invented for his steam digester was an important technological advance for the use of steam as a motive power.

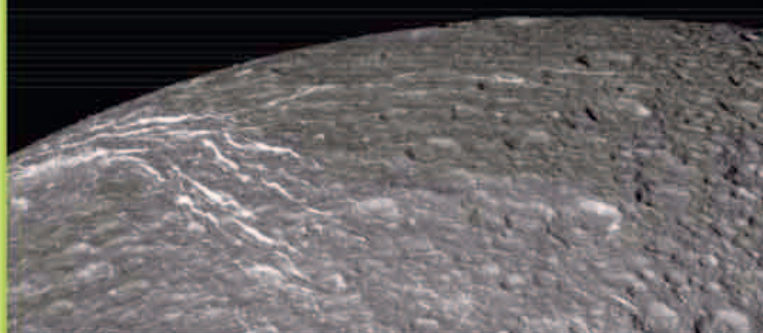


On the Motion of Animals
In this book, Giovanni Borelli applied the physical principles of mechanics to describe how the living body works and moves.

his career studying the movements of animals. He realized that **muscle contraction** relied on chemical processes and nervous stimulation. His pioneering work in this new field of biomechanics was published in 1680, a year after his death.



Boiling points
Water boils at higher temperatures as pressure is raised. As a result, food cooks at higher temperatures in boiling water under pressure.



Dione, one of Saturn's moons, was discovered by Italian astronomer Giovanni Cassini in 1684.

IN 1682, AS ENGLISH ASTRONOMER EDMOND HALLEY plotted the orbit of a comet, he realized that its characteristics matched those of comets recorded in 1531 and 1607. He deduced that they were all the same comet—which today bears his name.

Also this year, English botanist **Nehemiah Grew** (1641–1712) published his book *The Anatomy of Plants*, one of the earliest comprehensive texts on **plant biology**. Grew often collaborated with the Italian **Marcello Malpighi** on **microscopic**

Giovanni Cassini had been studying the planet Saturn, and by 1684 discovered four of its moons. He called them *Sidera Lodoicea*—the **Stars of Louis**—in honor of Louis XIV, patron of the Paris Observatory. Individually, the moons are now named Iapetus, Rhea, Dione, and Tethys.

Robert Hooke and Edmond Halley had been collaborating on trying to explain observed **planetary motions** based on mathematical laws that had been described by German astronomer **Johannes Kepler** at the start of the century. When

“ **ALL NATURE IS AS ONE GREAT ENGINE, MADE BY, AND HELD IN HIS HAND.** ”

Nehemiah Grew, English botanist, in *The Anatomy of Plants*, 1682

anatomy—Grew concentrated on plants, Malpighi on animals. Previously, Grew had extracted the green plant pigment—called **chlorophyll** today—and may have made some of the earliest observations on chloroplasts. He also asserted that plants reproduce sexually (in other words, have male and female parts), and found that pollen grains had distinctive surface sculpturing (see 1916–17).

they were unable to do so, in 1684, Halley visited English physician and mathematician **Isaac Newton** in Cambridge to gauge his opinion, only to be told that Newton had already resolved the issue. Encouraged by Halley, Newton went on to explain the **elliptical orbits of planets**, which he eventually incorporated in *Principia* (see 1687–89).

1679 Gottfried Leibniz describes a system of binary arithmetic

1679 Denis Papin demonstrates a steam digester

1680 Giovanni Borelli explains the mechanics of the living body

September 1682 Halley plots the course of the comet that will later carry his name

1682 Nehemiah Grew publishes *Anatomy of Plants*

1684 Cassini discovers four of Saturn's moons

December 10, 1684 Isaac Newton explains planetary motion based on Kepler's laws